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## ARTICLE INFO

## JEL classification:

C22

## Keywords:

Interest rate forecast

Predictive regression

Macroeconomic trend

Shifting endpoints

Trend estimation

## ABSTRACT

Treasury yields in the global bond market exhibit a secular decline in the past four decades. We show that this long-run trend of yield curve is associated with two key macroeconomic variables, the trend inflation and the equilibrium real short rate. These variables demonstrate substantial variation over time. Accounting for time variation of these macro trends is crucial for understanding yield dynamics. Changes in both trends can explain high persistence in interest rates. Furthermore, substantial predictive gain is obtained by including both trends in the predictive regression for bond risk premia in international bond markets.

## 1. Introduction

Over the past decades, numerous attempts have been made to investigate the link between macroeconomic variables and the term structure of interest rates. Although there is consensus that macroeconomic fundamentals are crucial for explaining yield curve dynamics, how to incorporate these fundamentals into analysis remains inconclusive. One particular attention is paid to explore the long-run means between macroeconomic variables and interest rates. Modern finance theory indicates that two key macroeconomic trends, the trend in inflation ( $\pi_t^*$ ) and the trend or equilibrium real short rate ( $r_t^*$ ), must be reflected in interest rates. Recently [Bauer and Rudebusch \(2020\)](#) confirm that there is significant variation in these long-run trends, and the secular decline in US long-term Treasury yields since the early 1980s can be largely explained by the movements in these underlying macroeconomic trends.

In this paper, we find that long-run trends of both variables have experienced substantial variation over time globally. We show that Treasury yields in the major bond markets, such as Canada, Germany, Japan, and UK, exhibit a secular decline in the long run.<sup>1</sup> We then demonstrate that variation in the trend inflation and the equilibrium real short rate, and the resulting nominal rate, which is the

<sup>☆</sup> We would like to thank Samuel Vigne (the editor), an anonymous referee, and seminar participants at Shanghai University for helpful comments. Yugui Zhang acknowledges the financial support from the Major Projects of Scientific Research and Innovation in Humanities and Social Sciences of Shanghai Municipal Education Commission (Grant no. 2017-01-07-00-06-E00031). Jie Zhu acknowledges the financial support from the Humanities and Social Science Planning Fund, Ministry of Education of China (Grant no. 19YJA790128). Xiaoneng Zhu acknowledges the financial support from the National Social Science Foundation (Grant no. 20&ZD102), SUFE (Grant no. 2018110698), and SIIFE (Grant no. 2018110262). All remaining errors are our own.

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<sup>1</sup> [Wu \(2020\)](#) provides theoretical insights and article reviews for the downward trend of real interest rates in major developed economies since the 1980s.

sum of them, can explain the high persistence in interest rates in these markets. Furthermore, incorporating changes in these trends is crucial for predicting bond risk premia in all markets. We show that adding these trends as regressors in the canonical predictive regressions for excess bond returns provides notable gains in predictive power, compared to those only including current yields (Fama and Bliss, 1987; Campbell and Shiller, 1991; Cochrane and Piazzesi, 2005).

Our work contributes to the literature in twofold. We make the first attempt to link the long-run trend in macroeconomic variables to the term structure of interest rates in international bond markets. Bauer and Rudebusch (2020) incorporate both macroeconomic trends into yield curve analysis, but they only consider the US market. We complement their study by extending the analysis to other markets. We find that both the trend inflation and the equilibrium real short rate are essential for understanding bond risk premia in all four markets. The two drivers for explaining the declining Treasury yields evolve differently over time in these markets. This fact implies that  $\pi_t^*$  and  $r_t^*$  may affect interest rates in different ways across markets. By investigating their impact on interest rates in the international bond market, we can better understand the economic link between these macro variables and interest rates.

Secondly, we provide new evidence on the macro-spanning puzzle from an international bond market perspective. The spanning hypothesis indicates that the current yield curve contains all relevant information for predicting future interest rates. However, the empirical evidence remains inconclusive. A number of researchers (e.g., Duffee, 2013; Bauer and Rudebusch, 2017; Bauer and Hamilton, 2018) find that macro variables contain little information for predicting excess bond returns. But other studies (e.g., Ludvigson and Ng, 2009; Chernov and Mueller, 2012; Cieslak and Povala, 2015) show that macro factors contain information for future bond risk premia, which is unspanned in the current yield curve. While most existing studies focus on the US market, we address the research question on spanning hypothesis by providing the international evidence. We show that macro factors, such as  $\pi_t^*$  and  $r_t^*$ , contain useful information for predicting bond risk premia in major bond markets. The paper is also related to a strand of literature on international bond markets (Dahlquist and Hasseltoft, 2013; Zhu, 2015; Hong et al., 2019; Zhao et al., 2020), which shows that global yield factors can predict international bond returns. We demonstrate that macro trends play an important role for predicting excess bond returns in these markets.

The remainder of this paper is organized as follows. Section 2 discusses how to estimate the trend inflation and equilibrium real short rate. It also presents the predictive regression including these macro trends. Section 3 describes the data and reports the empirical findings. Section 4 concludes the paper.

## 2. Methodology

### 2.1. Interest rate trends and macro trends

We start with the usual decomposition for any  $n$ -period bond yield at time  $t$ ,  $y_{nt}$ , into expectations of short-term rates and a term premium,  $TP_{nt}$ :

$$y_{nt} = \frac{1}{n} \sum_{j=0}^{n-1} E_t i_{t+j} + TP_{nt} = i_t^* + \frac{1}{n} \sum_{j=0}^{n-1} E_t i_{t+j}^c + TP_{nt}, \quad (1)$$

where  $i_t$  is the nominal short rate,  $y_{1t}$ . Eq. (1) indicates that any  $n$ -period long-term bond yield should equal to the averaged future short rate plus a term premium. The term premium compensates bond investors for holding longer-term bonds. It may also capture duration risk, liquidity risk, or financial market frictions. The traditional expectation theory of term structure in interest rates assumes that the term premium is either zero (strong form) or constant (weak form), while time variation in the term premium is now acknowledged as a stylized fact in interest rates. By introducing the equilibrium or the trend short rate,  $i_t^*$ , and short rate cycle,  $i_t^c$ , we may write  $i_t$  as the sum of  $i_t^*$  and  $i_t^c$ , and thus the second equality in Eq. (1) appears.  $i_t^*$  is defined as the limiting conditional short-term nominal interest rate:

$$i_t^* = \lim_{j \rightarrow \infty} E_t i_{t+j}. \quad (2)$$

We assume that  $i_t^*$  is a nonstationary stochastic trend. Eq. (1) indicates that all nominal yields are affected by changes in  $i_t^*$ . This common trend implies that yields are nonstationary if  $i_t^*$  is time varying. In this sense,  $i_t^*$  is called a "shifting endpoint". The Fisher equation indicates that  $i_t^*$  is determined by two macroeconomic trends:

$$i_t = r_t + E_t \pi_{t+1} \Rightarrow i_t^* = r_t^* + \pi_t^*, \quad (3)$$

where  $r_t$  is the real short rate and  $\pi_t$  is the inflation rate.  $r_t^*$  and  $\pi_t^*$  are the trends in real short rate and inflation, respectively. The nominal interest rates are driven by both macroeconomic trends. Consequently all bond yields are affected by these trends.

Since neither  $\pi_t^*$  nor  $r_t^*$  is observable, we have to estimate these variables from data. To avoid the problem that historical macroeconomic time series are often subject to significant revisions after initial releases and to keep the estimation procedure as simple as possible, we measure the trend inflation based on a discounted moving average (DMA) of past core inflation, which allows people to gradually update their expectations to inflation over time. As illustrated in Cieslak and Povala (2015), such a measure can indeed reflect future inflation quite well. Specifically, the DMA of past inflation is constructed as follows

$$\pi_t^* = (1 - \nu) \sum_{i=0}^{t-1} \nu^i \pi_{t-i}, \quad (4)$$

where  $\pi_t = \ln \frac{CPI_t}{CPI_{t-1}}$  is the quarterly-sampled inflation rate. Following previous literature, we truncate the sum in Eq. (4) at  $N = 120$  months and choose  $\nu = 0.987$ .<sup>2</sup>

As a proxy for  $r_t^*$ , we use a simple exponentially-weighted moving average of past real rates.

$$r_t^* = \alpha r_{t-1}^* + (1 - \alpha) r_t. \quad (5)$$

Here  $r_t$  is the observed real short rate at  $t$ . Following earlier literature in estimating macro trends, we set the value of  $\alpha$  to be 0.98 and the initial value of  $r_0^* = r_0$ . We notice it is a model-free and real-time estimate of  $r_t^*$ . As shown in Bauer and Rudebusch (2020), this measure is highly correlated with other model-based estimates. Note that estimates for both  $\pi_t^*$  and  $r_t^*$  are real-time. In that way, it avoids the problem for possible bias bringing by revised macro data.

## 2.2. Cointegration between yields and macro trends

It is well documented that interest rates are highly persistent. We first investigate whether variation in the inflation trend and the real short rate trend contributes to the highly persistent movements in nominal yields. To do that, we construct the detrended yield series by differencing the yield and relative macro trends, i.e.  $\pi_t^*$  or  $r_t^*$ . To see whether accounting for just the inflation trend is sufficient or if a combination of the inflation trend and the real short rate trend contributes more to the movements in nominal yield, we also construct the detrended yield series by differencing the yield and  $i_t^*$ . To do that, we run the cointegration regression of the yield on the macro trend. The detrended yield should be stationary if the yield is indeed driven by the macro trend and the loading on the macro trend should be unity. Also the residual from the cointegration regression of the yield on the macro trend should be less persistent if the macro trend really captures the high persistence in yields.

## 2.3. Predictive regression for excess bond returns

A natural empirical benchmark for judging the predictive ability of macroeconomic trends is a predictive regression including yield curve factors and these trends. Previous literature documents that the yield curve contains useful information for predicting excess bond returns, i.e. Fama and Bliss (1987) and Campbell and Shiller (1991).

Define the one-quarter return on a  $n$ -period bond in excess of the risk-free rate as follows

$$rx_{n,t+1} = p_{n-1,t+1} - p_{nt} - y_{1t} = -(n-1)y_{n-1,t+1} + ny_{nt} - y_{1t}, \quad (6)$$

where  $p_{nt}$  denotes the log price of a zero-coupon bond.<sup>3</sup> Since previous literature shows that the first three principal components (PCs) from the current yield curve – level, slope, and curvature – help to predict future yields, we include these PCs in the predictive regression. In addition, we also include macroeconomics trends,  $\pi_t^*$ ,  $r_t^*$ , or  $i_t^*$ , in the regression to see whether these trends are useful in forecasting excess bond returns. The predictive regression is expressed as follows

$$rx_{n,t+1} = \alpha + A' PC_t + B' x_t + \varepsilon_{t+1}, \quad (7)$$

where  $PC_t$  is a vector including three PCs at  $t$  and  $x_t$  is a vector including macro trends, and  $A$  and  $B$  are corresponding parameter vectors.

## 3. Empirical analysis

### 3.1. Data and summary statistics

We consider four major bond markets in this paper: Canada, Germany, Japan, and United Kingdom. We choose these markets for several reasons. These markets are the largest in the world only behind US. These markets are very liquid and bond returns are less affected by illiquidity. Also a panel of real-time macro variables are available for these markets. Finally all foreign exchange risk can be hedged in these markets.

The data of interest rates is made up of end-of-quarter observations of the 3-month to the longest maturity zero-coupon Treasury bond yields.<sup>4</sup> We choose the quarterly data because some macro variables are only available at quarterly frequency. The samples start

<sup>2</sup> A sensitivity analysis in Cieslak and Povala (2015) shows that varying  $N$  between 100 and 150 months and  $\nu$  between 0.975 and 0.995 leads to very small quantitative differences in their empirical analysis.

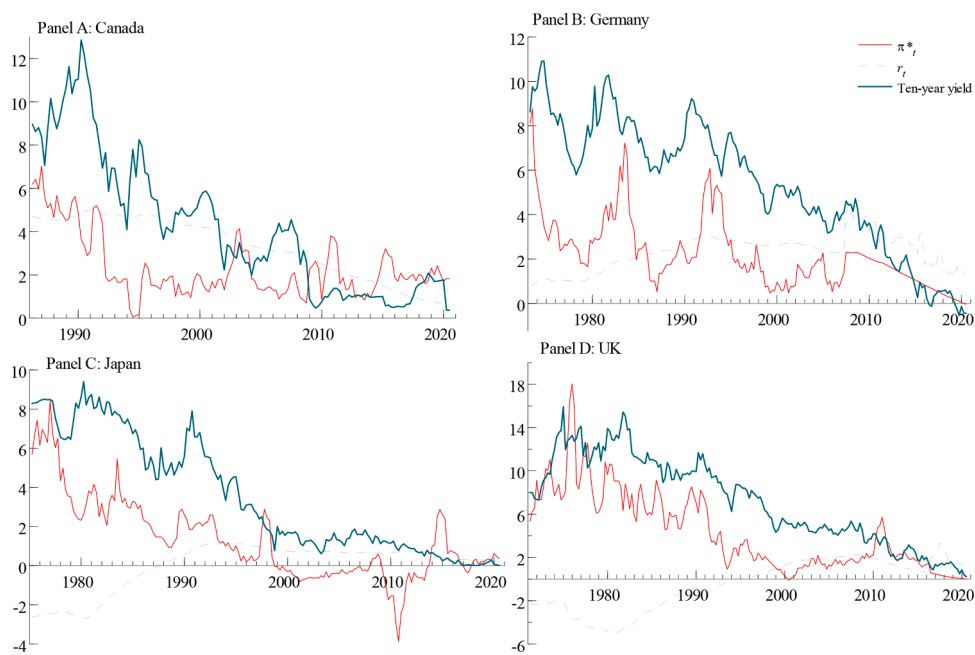
<sup>3</sup> Similar to Bauer and Rudebusch (2020), we choose the one-quarter holding periods to avoid overlapping observations. We use the usual approximation  $y_{n-1,t+1} \approx y_{n,t}$  so that Eq. (6) actually becomes  $rx_{n,t+1} = y_{nt} - y_{1t}$ .

<sup>4</sup> The longest maturity of zero-coupon Treasury bond yield varies across markets and time. It is because we attempt to include bond yields as many as possible.

**Table 1**  
Summary statistics for bond yields and trend estimates.

|                                            | Canada  | Germany | Japan  | UK      |
|--------------------------------------------|---------|---------|--------|---------|
| Start Date                                 | 1986Q1  | 1973Q1  | 1975Q1 | 1971Q1  |
| End Date                                   | 2020Q2  | 2020Q2  | 2020Q2 | 2020Q2  |
| Range of Bond Maturities (years)           | 0.25–25 | 1–30    | 1–40   | 0.50–40 |
| Mean of 10-year bond yields (%)            | 5.26    | 5.33    | 2.25   | 7.26    |
| Mean of averaged long-term bond yields (%) | 5.35    | 5.29    | 2.10   | 7.26    |
| Mean of $\pi_t^*$ (%)                      | 2.33    | 2.26    | 0.40   | 4.49    |
| Mean of $r_t^*$ (%)                        | 3.29    | 1.90    | 0.65   | −0.62   |
| Mean of $i_t^*$ (%)                        | 5.62    | 4.16    | 1.05   | 3.87    |

*Note:* This table reports summary statistics for bond yields and trend estimates used in the paper.  $\pi_t^*$  is the trend estimate for inflation,  $r_t^*$  is the trend estimate for the real short rate, and  $i_t^*$  is the trend estimate for the short-term nominal interest rate. The averaged long-term bond yields indicate the averaged yields in percentage for all bonds with maturities equal or longer than two years in each period. The sample size varies from 138 quarters (Canada) to 198 quarters (UK). All samples are quarterly observations.



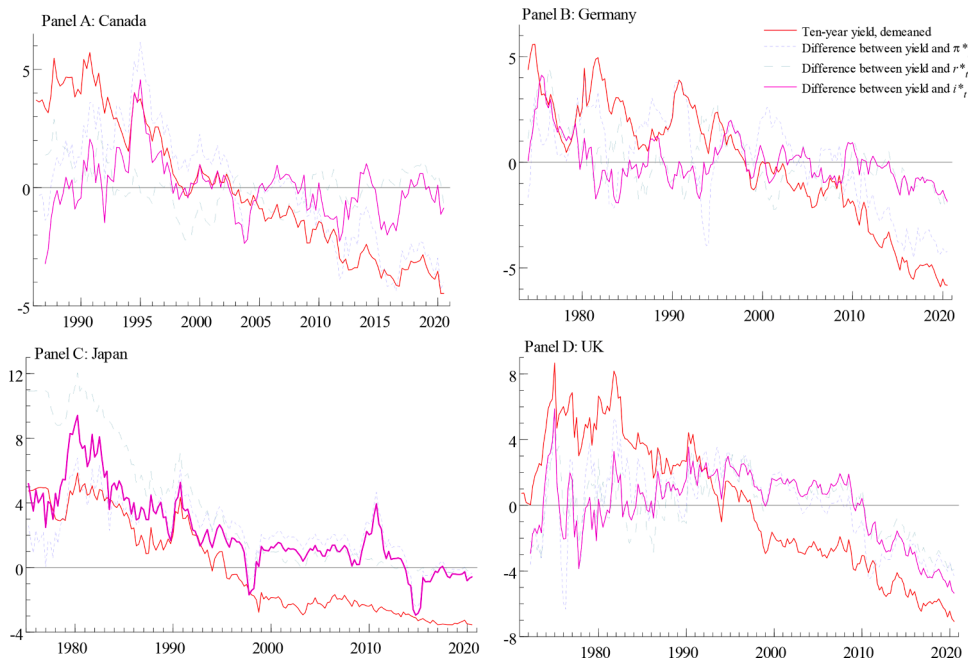
**Fig. 1.** Ten-year yields and macroeconomic trends. Ten-year Treasury yield and estimates of the trend inflation,  $\pi_t^*$ , and the equilibrium real rate,  $r_t^*$ . The data start from various dates for different markets and end at 2020 : Q2.

differently across markets and end at 2020Q2. We collect the Canada yield data from Bank of Canada, the Germany yield data from the Bundesbank, the central bank of Germany, the Japan yield data from the Ministry of Finance in Japan, and the UK yield data from the Bank of England. The macro variables for these markets are obtained from the OECD dataset.

Table 1 reports the sample period, the range of maturities, and summary statistics of Treasury bond yields for the four markets. For most markets, the sample period starts from early 1970s, while for Canada it starts from 1986. The longest maturities for bond yields range from 25 (Canada) to 40 years (Japan and UK). While Japan has the minimum mean for long-term bond yields (both for the 10-year yield and averaged long-term bond yields ranging from 2 years to 40 years), UK experiences the highest mean of 7.26 in both cases. Not surprisingly, Japan also has the minimum mean of 0.40 for  $\pi^*$ , while UK has the minimum mean of −0.62 for  $r^*$ .

### 3.2. Empirical results

Fig. 1 shows time series dynamics of ten-year Treasury yields,  $\pi_t^*$ , and  $r_t^*$ . The secular decline in the yield is apparent for all markets.



**Fig. 2.** Detrending the ten-year yield. The four detrended ten-year Treasury yield series using the proxy for the trend inflation  $\pi_t^*$ , the equilibrium real interest rate  $r_t^*$ , and the equilibrium nominal interest rate  $i_t^*$ . The first detrended series is the demeaned yield, and the other three are the regression residuals between the yield and  $\pi_t^*$ ,  $r_t^*$ , and  $i_t^*$ , respectively. The data start from various dates for different markets and end at 2020 : Q2.

The long-run tendency of decline for the trend inflation is also notable, although it experiences some temporary rises. The dynamics of  $r_t^*$  also show a long-run decline, especially for Canada and Germany and in later periods, although the decline tendency is not apparent for Japan and UK in earlier periods. In order to show whether macro trends account for the persistence of interest rates, we plot the demeaned ten-year Treasury yield and three detrended versions of that yield in Fig. 2.<sup>5</sup> The demeaned yield exhibits an apparent trend of decline in the long-run for all markets, while the three detrended series show less of such a trend. For example, take a look of Panel A for the case of Canada, the purple line plots the detrended yield with the difference between the yield and  $i_t^*$ . We can see while this line is below the yield in the early sample, it moves over the yield after 2010. It shows much less of a trend than the demeaned yield, which indicates that  $i_t^*$  can indeed capture the long-run trend in the yield. The similar result holds for  $\pi_t^*$  and  $r_t^*$ , and also for other markets.

Table 2 presents the results for the cointegration regression. The table reports the parameter estimates, Newey-West standard errors, and the regression  $R^2$ . It also provides Augmented Dickey-Fuller (ADF) unit root tests and 1-lag autocorrelation for the cointegration residual.

The first column of Table 2 presents the results of the regression only including a constant, which confirms the high persistence in interest rates. The residual is highly autocorrelated and the ADF test for the residual doesn't reject the null. Including  $\pi_t^*$  leads to a higher  $R^2$ , a lower standard deviation, a lower autocorrelation, and the coefficient on  $\pi_t^*$  is not far from 1. Adding  $r_t^*$  in the regression keeps the tendency, i.e. it can raise  $R^2$  further with a lower standard deviation, and now the ADF test for the residual significantly rejects the null. It seems that only including  $\pi_t^*$  is not sufficient, and adding  $r_t^*$  improves the detrending for yields. Note the coefficient of  $r_t^*$  is negative for Japan and UK. It indicates that  $r_t^*$  exerts different impact across markets. Fig. 1 shows that  $r_t^*$  tends to move opposite to Treasury yields in some periods in Japan, and UK. This leads to the negative coefficient of  $r_t^*$  on Treasury yields in these two markets. The coefficient of the nominal interest rate,  $i_t^*$ , is significantly positive for all markets. It implies that although  $r_t^*$  has negative impact on Treasury yields in Japan and UK, it is less important in affecting Treasury yields. The regression  $R^2$  is a little lower than that for the regression including both  $\pi_t^*$  and  $r_t^*$ , but higher than the  $R^2$  for the regression including either  $\pi_t^*$  or  $r_t^*$  alone. It seems that  $\pi^*$  is not sufficient for detrending yields, and  $r^*$  plays a more important role in doing that.

We now run the predictive regression to investigate whether macroeconomic trends can help forecast future bond excess returns. It is well known that the yield curve contains useful information for predicting excess bond returns. In that spirit, we include the first three principal components (PCs) in the regression. We then add the inflation trend, the real short rate trend, and the sum of these trends one by one in the regression. Whether these trends can add predictive power to the PCs is the key question we attempt to answer. Some previous studies, such as Cieslak and Povala (2015) and Bauer and Hamilton (2018) find the inflation trend is a robust predictor and is unspanned by the current yield curve. We analyze the issue more broadly by considering  $r_t^*$  and  $i_t^*$  in addition to  $\pi_t^*$ .

<sup>5</sup> We regress the ten-year Treasury yields on current and lagged  $\pi^*$ ,  $r^*$ , or  $i^*$ , respectively. The regression residuals are used for plotting the detrended series.

**Table 2**  
Cointegration regressions and tests for the level of the yield curve.

| Panel A: Canada |                   |                    |                     |                    | Panel B: Germany |                   |                    |                     |                    |
|-----------------|-------------------|--------------------|---------------------|--------------------|------------------|-------------------|--------------------|---------------------|--------------------|
|                 | Yield             | (1)                | (2)                 | (3)                |                  | Yield             | (1)                | (2)                 | (3)                |
| Constant        | 5.26***<br>(0.24) | 3.02***<br>(0.40)  | − 1.46***<br>(0.22) | − 0.78**<br>(0.32) | Constant         | 5.33***<br>(0.21) | 2.66***<br>(0.31)  | 0.073<br>(0.44)     | 0.20<br>(0.40)     |
| $\pi_t^*$       |                   | 0.96***<br>(0.15)  | 0.46***<br>(0.058)  |                    | $\pi_t^*$        |                   | 1.18***<br>(0.11)  | 1.20***<br>(0.10)   |                    |
| $r_t^*$         |                   |                    | 1.72***<br>(0.060)  |                    | $r_t^*$          |                   |                    | 1.34***<br>(0.18)   |                    |
| $i_t^*$         |                   |                    |                     | 1.07***<br>(0.053) | $i_t^*$          |                   |                    |                     | 1.23***<br>(0.089) |
| $R^2$           | 0.77              | 0.84               | 0.89                | 0.87               | $R^2$            | 0.30              | 0.36               | 0.50                | 0.50               |
| SD              | 2.85              | 2.47               | 0.94                | 1.41               | SD               | 2.91              | 2.32               | 2.05                | 2.08               |
| $\hat{\rho}$    | 0.97              | 0.93               | 0.80                | 0.79               | $\hat{\rho}$     | 0.97              | 0.92               | 0.89                | 0.91               |
| ADF             | − 1.59            | − 2.63*            | − 3.95***           | − 3.21**           | ADF              | − 1.74            | − 3.27**           | − 3.37**            | − 3.48**           |
| Panel C: Japan  |                   |                    |                     |                    | Panel D: UK      |                   |                    |                     |                    |
|                 | Yield             | (1)                | (2)                 | (3)                |                  | Yield             | (1)                | (2)                 | (3)                |
| Constant        | 3.55***<br>(0.22) | 2.02***<br>(0.15)  | 2.48***<br>(0.18)   | 1.94***<br>(0.25)  | Constant         | 7.26***<br>(0.29) | 3.22***<br>(0.29)  | 5.01***<br>(0.32)   | 3.99***<br>(0.54)  |
| $\pi_t^*$       |                   | 1.15***<br>(0.063) | 0.81***<br>(0.092)  |                    | $\pi_t^*$        |                   | 0.90***<br>(0.051) | 0.36***<br>(0.077)  |                    |
| $r_t^*$         |                   |                    | − 0.75***<br>(0.16) |                    | $r_t^*$          |                   |                    | − 1.05***<br>(0.12) |                    |
| $i_t^*$         |                   |                    |                     | 1.20***<br>(0.13)  | $i_t^*$          |                   |                    |                     | 0.85***<br>(0.12)  |
| $R^2$           | 0.58              | 0.65               | 0.68                | 0.67               | $R^2$            | 0.53              | 0.61               | 0.72                | 0.70               |
| SD              | 2.96              | 1.75               | 1.65                | 1.71               | SD               | 4.03              | 2.50               | 2.13                | 2.26               |
| $\hat{\rho}$    | 0.98              | 0.89               | 0.85                | 0.86               | $\hat{\rho}$     | 0.98              | 0.86               | 0.81                | 0.83               |
| ADF             | − 1.09            | − 2.91**           | − 3.42**            | − 3.15**           | ADF              | − 1.16            | − 2.62*            | − 3.17**            | − 2.95**           |

Note: Dynamic OLS regressions for the ten-year yield,  $y_t^{(10)}$ , on macroeconomic trends. Newey-West standard errors using six lags are in parenthesis. The  $\pi_t^*$  and  $r_t^*$  are the trend inflation and the equilibrium real short rate as described in Section 2, and the long-run nominal short rate,  $i_t^*$ , is the sum of  $\pi_t^*$  and  $r_t^*$ . For the cointegration residuals, the table reports standard deviations (SD), first-order autocorrelation coefficients ( $\hat{\rho}$ ), and Augmented Dickey-Fuller (ADF) unit root test statistics. \*\*\*, \*\*, and \* indicate significant level at 1%, 5%, and 10%, respectively.

We forecast the one-quarter long-term excess return averaged across maturities from 2 years to the maturity as long as possible. Table 3 reports the results for four different predictive regressions for the full sample, and Table 4 reports the results for a shorter subsample starting from 1990Q1 for robustness.<sup>6</sup> Column (1) serves as a baseline case, which includes a constant and three PCs. Columns (2) to (4) add  $\pi_t^*$ ,  $r_t^*$ , and  $i_t^*$  to the regression sequentially. For the full sample, Table 3 shows that including the inflation trend can obtain substantial gains in predicting bond returns. For the case of Canada, it raises  $R^2$  from 0.08 to 0.12. Thus it seems that  $\pi_t^*$  contains some information not spanned by the yield curve. But  $\pi_t^*$  alone is not sufficient. By adding  $r_t^*$  we can raise  $R^2$  further to 0.17. Since  $i_t^*$  is the sum of  $\pi_t^*$  and  $r_t^*$ , it should carry relevant information in both trends, which is confirmed in the regression (4). The regression  $R^2$  for including  $i_t^*$  is 0.16, which is similar to the one in (3) with  $r_t^*$  and  $\pi_t^*$ . Across markets, the coefficient on  $\pi_t^*$  is negative in most cases. However, the coefficient of  $r_t^*$  is positive for Canada and Germany, albeit it is negative for Japan and UK. It implies that  $r_t^*$  has asymmetric impact on the expectations of short rates and spot short rate.<sup>7</sup> For example, Table 2 shows that  $r_t^*$  keeps relatively low in Japan and UK. If investors expect that interest rates tend to be low in the long term, then an increase in  $r_t^*$  will immediately lead to increase in  $y_{1t}$ , but with a less increase in expectations of short-term rates. This indicates a decrease in the bond excess returns. These results confirm that adding  $\pi_t^*$ ,  $r_t^*$ , or  $i_t^*$  can improve the predictability. The extra predictive gain from  $r_t^*$  or  $i_t^*$  is not negligible. The regression  $R^2$  is raised considerably by adding these variables across markets.

We obtain similar results from the subsample. Most regression  $R^2$  in columns (1), (2), (3), and (4) of Table 4 are higher than their counterparts in the full sample. We find that most coefficients of  $\pi_t^*$  are negative both in the full sample and the subsample. This implies that the trend inflation has asymmetric impact on long-term bond yield and short rate. That is, if investors believe that the inflation will increase in the future, they require a higher yield for the long-term bond, thus the long-term bond yield will increase more than the

<sup>6</sup> We choose the subsample to start from 1990Q1 since there is some evidence that yields and macro trends experience a shift in early 1990s, as shown in Fig. 1.

<sup>7</sup> Combining Eqs. (1) and (6), and use the usual approximation  $y_{n-1,t+1} \approx y_{nt}$ , we can see that the bond excess return,  $rx_{t+1} = \frac{1}{n} \sum_{j=0}^{n-1} E_t i_{t+j} + TP_{nt} - y_{1t}$ .  $r_t^*$  has impact on  $\frac{1}{n} \sum_{j=0}^{n-1} E_t i_{t+j}$  and  $y_{1t}$ .

**Table 3**

Predictive regressions: yields and macro trends, full sample.

|                                   | (1)                    | (2)                    | (3)                    | (4)                    |
|-----------------------------------|------------------------|------------------------|------------------------|------------------------|
| Panel A: Canada, 1986:Q1–2020:Q2  |                        |                        |                        |                        |
| PC1                               | 0.12***<br>(0.017)     | 0.06***<br>(0.018)     | 0.11**<br>(0.050)      | 0.013<br>(0.034)       |
| PC2                               | 0.13***<br>(0.0049)    | 0.12***<br>(0.0051)    | 0.12***<br>(0.0050)    | 0.13***<br>(0.0046)    |
| PC3                               | – 0.022***<br>(0.0081) | – 0.043***<br>(0.0084) | – 0.045***<br>(0.0085) | – 0.035***<br>(0.0082) |
| $\pi_t^*$                         |                        | – 0.21***<br>(0.040)   | – 0.20***<br>(0.042)   |                        |
| $r_t^*$                           |                        |                        | 0.099**<br>(0.042)     |                        |
| $i_t^*$                           |                        |                        |                        | – 0.18***<br>(0.043)   |
| $R^2$                             | 0.08                   | 0.12                   | 0.17                   | 0.16                   |
| Panel B: Germany, 1973:Q1–2020:Q2 |                        |                        |                        |                        |
| PC1                               | – 0.10***<br>(0.019)   | – 0.071***<br>(0.024)  | – 0.11***<br>(0.030)   | – 0.075**<br>(0.028)   |
| PC2                               | 0.13***<br>(0.0049)    | 0.022<br>(0.029)       | – 0.073<br>(0.051)     | – 0.053<br>(0.051)     |
| PC3                               | – 0.022***<br>(0.0081) | – 0.066<br>(0.047)     | – 0.11**<br>(0.050)    | – 0.11**<br>(0.050)    |
| $\pi_t^*$                         |                        | – 0.14**<br>(0.065)    | – 0.14**<br>(0.064)    |                        |
| $r_t^*$                           |                        |                        | 0.43**<br>(0.19)       |                        |
| $i_t^*$                           |                        |                        |                        | – 0.087**<br>(0.041)   |
| $R^2$                             | 0.15                   | 0.18                   | 0.24                   | 0.23                   |
| Panel C: Japan, 1975:Q1–2020:Q2   |                        |                        |                        |                        |
| PC1                               | 0.021<br>(0.017)       | 0.10***<br>(0.031)     | – 0.059<br>(0.045)     | 0.078***<br>(0.021)    |
| PC2                               | – 0.12***<br>(0.049)   | – 0.15***<br>(0.030)   | 0.10*<br>(0.060)       | – 0.079***<br>(0.029)  |
| PC3                               | – 0.078***<br>(0.0029) | – 0.078***<br>(0.0028) | – 0.071***<br>(0.0030) | – 0.075***<br>(0.0028) |
| $\pi_t^*$                         |                        | 0.16***<br>(0.031)     | – 0.23***<br>(0.050)   |                        |
| $r_t^*$                           |                        |                        | – 0.73***<br>(0.16)    |                        |
| $i_t^*$                           |                        |                        |                        | – 0.23***<br>(0.051)   |
| $R^2$                             | 0.08                   | 0.10                   | 0.18                   | 0.16                   |
| Panel D: UK, 1971:Q1–2020:Q2      |                        |                        |                        |                        |
| PC1                               | 0.32***<br>(0.059)     | 0.086<br>(0.47)        | – 0.063<br>(0.10)      | 0.28***<br>(0.068)     |
| PC2                               | 0.11***<br>(0.032)     | 0.10***<br>(0.031)     | 0.16***<br>(0.032)     | 0.094***<br>(0.034)    |
| PC3                               | – 0.21***<br>(0.032)   | – 0.19***<br>(0.032)   | – 0.18***<br>(0.032)   | – 0.20***<br>(0.032)   |
| $\pi_t^*$                         |                        | 0.31***<br>(0.10)      | 0.089**<br>(0.041)     |                        |
| $r_t^*$                           |                        |                        | – 0.67***<br>(0.23)    |                        |
| $i_t^*$                           |                        |                        |                        | 0.12***<br>(0.013)     |
| $R^2$                             | 0.29                   | 0.32                   | 0.38                   | 0.35                   |

Note: Predictive regressions for quarterly excess bond returns, averaged across 2 years to the longest maturity. The predictors are the first three principal components of yields (PC1, PC2, PC3), and estimates of the trend inflation,  $\pi_t^*$ , the real time equilibrium short rate,  $r_t^*$ , and the nominal short rate,  $i_t^*$ , which is the sum of  $\pi_t^*$  and  $r_t^*$ . The data are quarterly and start from various dates and end at 2020:Q2. \*\*\*, \*\*, and \* indicate significant level at 1%, 5%, and 10%, respectively.

**Table 4**

Predictive regressions: yields and macro trends, subsample, 1990:Q1–2020:Q2.

|                  | (1)                    | (2)                    | (3)                    | (4)                    |
|------------------|------------------------|------------------------|------------------------|------------------------|
| Panel A: Canada  |                        |                        |                        |                        |
| PC1              | 0.077**<br>(0.036)     | 0.081**<br>(0.034)     | 0.083<br>(0.061)       | 0.17***<br>(0.044)     |
| PC2              | 0.11***<br>(0.0049)    | 0.11***<br>(0.0046)    | 0.11***<br>(0.0049)    | 0.12***<br>(0.0048)    |
| PC3              | – 0.18***<br>(0.031)   | – 0.17***<br>(0.044)   | – 0.17***<br>(0.030)   | – 0.18***<br>(0.030)   |
| $\pi_t^*$        |                        | – 0.17***<br>(0.040)   | – 0.17***<br>(0.046)   |                        |
| $r_t^*$          |                        |                        | – 0.35***<br>(0.089)   |                        |
| $i_t^*$          |                        |                        |                        | – 0.15***<br>(0.045)   |
| $R^2$            | 0.10                   | 0.13                   | 0.19                   | 0.17                   |
| Panel B: Germany |                        |                        |                        |                        |
| PC1              | – 0.91***<br>(0.041)   | – 0.85***<br>(0.041)   | – 0.85***<br>(0.042)   | – 0.86***<br>(0.041)   |
| PC2              | 1.95***<br>(0.093)     | 1.87***<br>(0.090)     | 1.90***<br>(0.12)      | – 0.053<br>(0.051)     |
| PC3              | – 0.96***<br>(0.052)   | – 0.94***<br>(0.047)   | – 0.94***<br>(0.050)   | – 0.11**<br>(0.050)    |
| $\pi_t^*$        |                        | – 0.16***<br>(0.041)   | – 0.16***<br>(0.041)   |                        |
| $r_t^*$          |                        |                        | – 0.43***              |                        |
| $i_t^*$          |                        | (0.14)                 |                        | – 0.087**<br>(0.041)   |
| $R^2$            | 0.27                   | 0.30                   | 0.35                   | 0.33                   |
| Panel C: Japan   |                        |                        |                        |                        |
| PC1              | – 0.37***<br>(0.11)    | – 0.28**<br>(0.11)     | – 0.095<br>(0.13)      | 0.078***<br>(0.021)    |
| PC2              | 0.34***<br>(0.13)      | 0.27**<br>(0.13)       | – 0.043<br>(0.18)      | – 0.079***<br>(0.029)  |
| PC3              | – 0.024***<br>(0.0070) | – 0.028***<br>(0.0070) | – 0.029***<br>(0.0068) | – 0.075***<br>(0.0028) |
| $\pi_t^*$        |                        | – 0.088**<br>(0.034)   | – 0.069**<br>(0.034)   |                        |
| $r_t^*$          |                        |                        | 1.05**<br>(0.43)       |                        |
| $i_t^*$          |                        |                        |                        | – 0.23***<br>(0.051)   |
| $R^2$            | 0.06                   | 0.08                   | 0.15                   | 0.14                   |
| Panel D: UK      |                        |                        |                        |                        |
| PC1              | 0.23***<br>(0.053)     | 0.21***<br>(0.067)     | 0.22***<br>(0.073)     | 0.22***<br>(0.054)     |
| PC2              | 0.49***<br>(0.058)     | 0.49***<br>(0.060)     | 0.49***<br>(0.060)     | 0.49***<br>(0.059)     |
| PC3              | – 0.55***<br>(0.042)   | – 0.54***<br>(0.061)   | – 0.54***<br>(0.064)   | – 0.55***<br>(0.047)   |
| $\pi_t^*$        |                        | 0.48***<br>(0.16)      | 0.48***<br>(0.17)      |                        |
| $r_t^*$          |                        |                        | 0.17***<br>(0.018)     |                        |
| $i_t^*$          |                        |                        |                        | 0.35***<br>(0.12)      |
| $R^2$            | 0.41                   | 0.44                   | 0.49                   | 0.47                   |

Note: Predictive regressions for quarterly excess bond returns, averaged across 2 years to the longest maturity. The predictors are the first three principal components of yields (PC1, PC2, PC3), and estimates of the trend inflation,  $\pi_t^*$ , the real time equilibrium short rate,  $r_t^*$ , and the nominal short rate,  $i_t^*$ , which is the sum of  $\pi_t^*$  and  $r_t^*$ . The data are quarterly, and start from 1990:Q1 and end at 2020:Q2. \*\*\*, \*\*, and \* indicate significant level at 1%, 5%, and 10%, respectively.

short rate, making long-term bond excess return decrease. However, the impact of  $r_t^*$  on bond risk premium is less obvious. The change in  $r_t^*$  will affect both long-term and short-term bond yield, thus the impact of  $r_t^*$  on bond excess returns could be in either direction, depending on whether the long-term bond yield or the short rate is more affected by  $r_t^*$ . We also note that  $r_t^*$  is time varying. Fig. 1 shows that  $r_t^*$  experiences roughly inverse development in both Japan and UK. For example, in the early period (from 1975 to 1990) in Japan,  $r_t^*$  is negative, while in the later period, it is positive. This inverse relation implies that  $r_t^*$  may have different impact on interest rates for different periods. Finally we note although the sign for the coefficient of  $r_t^*$  shows to be opposite in the full sample and the subsample, the sum of  $\pi_t^*$  and  $r_t^*$ , i.e. the nominal short rate  $i_t^*$ , predicts bond excess returns consistently both in the full sample and the subsample.

#### 4. Conclusions

The apparent secular decline in long-term Treasury yields since 1980s is a well-observed fact in international bond markets. We show that this decline is related to shifting endpoints of macro trends. To investigate the issue, we explore the predictive power of macro trends for bond risk premia in different bond markets. We consider the trend inflation and the equilibrium real short rate. The real-time estimates for these trends are constructed. We implement the analysis for four major bond markets in the world: Canada, Germany, Japan, and UK. We find that allowing for persistent shifts in  $\pi_t^*$  and  $r_t^*$  is crucial for accounting for the high persistence in interest rates. Furthermore, the substantial predictive gain is obtained by including these trends in the predictive regression for excess bond returns. This paper supplements the study of macroeconomic variables on bond risk premia by providing international evidence. The result confirms the findings in Bauer and Rudebusch (2020) that slow-moving changes in the long-run means of nominal and real interest rates are related to macro trends. Future research avenues seem promising in this field.

#### Declaration of Competing Interest

None.

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